RESEARCH PRESENTATION

Artificial Intelligence Adoption in European Enterprises

A Multi-Level Analysis of Country, Firm Size and Sector Drivers, 2021–2025

GROUP X

Name One · Name Two · Name Three · Name Four

LUISS GUIDO CARLI • INTERNET ECONOMICS • MAY 2026

Why AI adoption — and why now?

The **Digital Decade Policy Programme 2030** set a target of 75% of EU enterprises using cloud, big data or AI by 2030. The AI KPI is the one flagged by the Commission as most likely to fall short.

Aggregate EU adoption **nearly tripled in four years** (7.7% → 20.7%). But the aggregate hides *three layers of inequality* that move in opposite directions.

This presentation asks one question: *where exactly is AI being adopted in Europe — and where is it not?*

Three numbers that frame the story

8×

ratio between AI adoption in Denmark (42%) and Romania (5%)

THE COUNTRY GAP

3×

large firms vs small firms in 2025 (54% vs 18%)

THE SIZE GAP

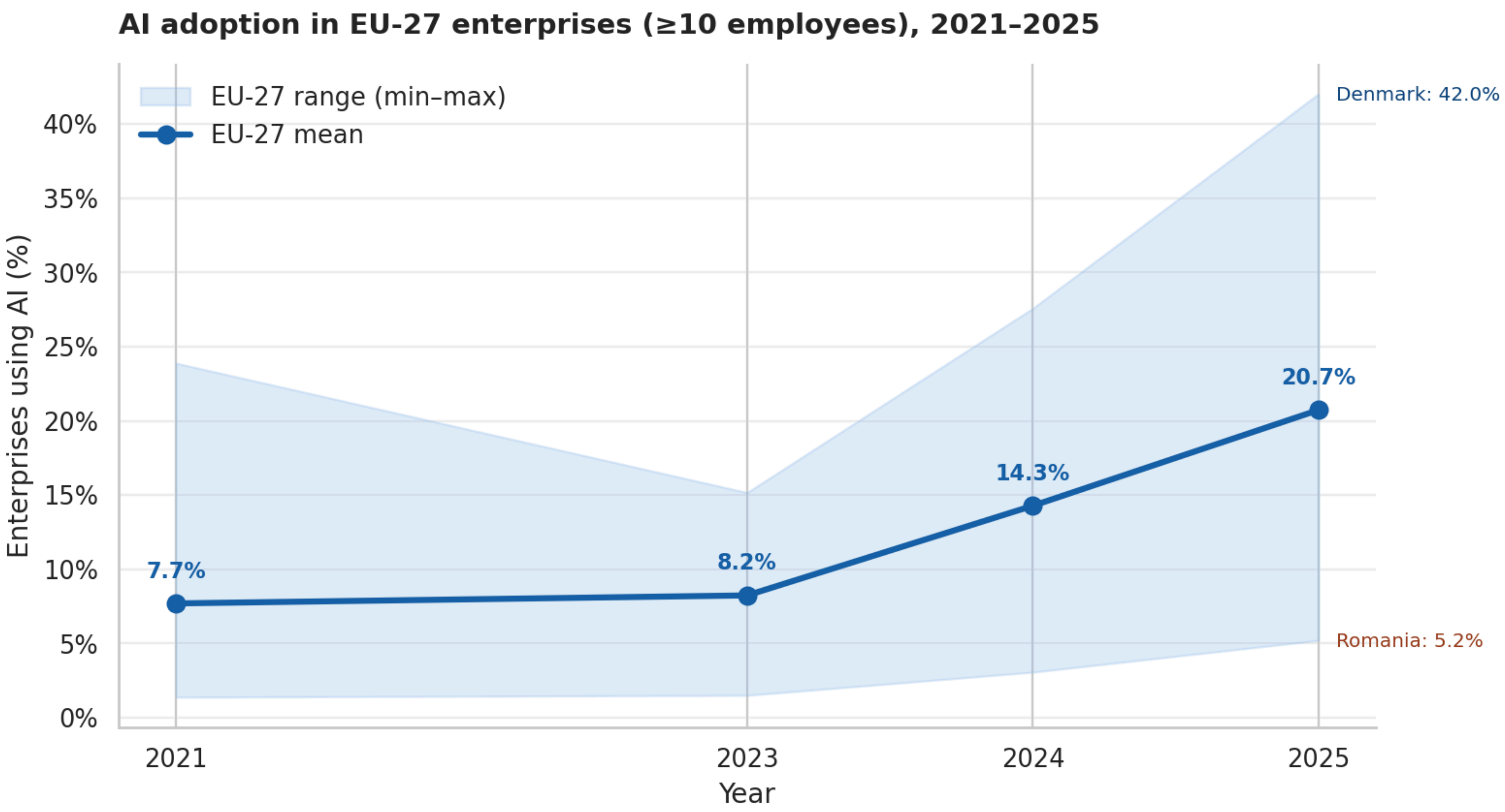
49pp

gap between ICT (60%) and construction (11%)

THE SECTOR GAP

No single layer explains the variance. They combine.

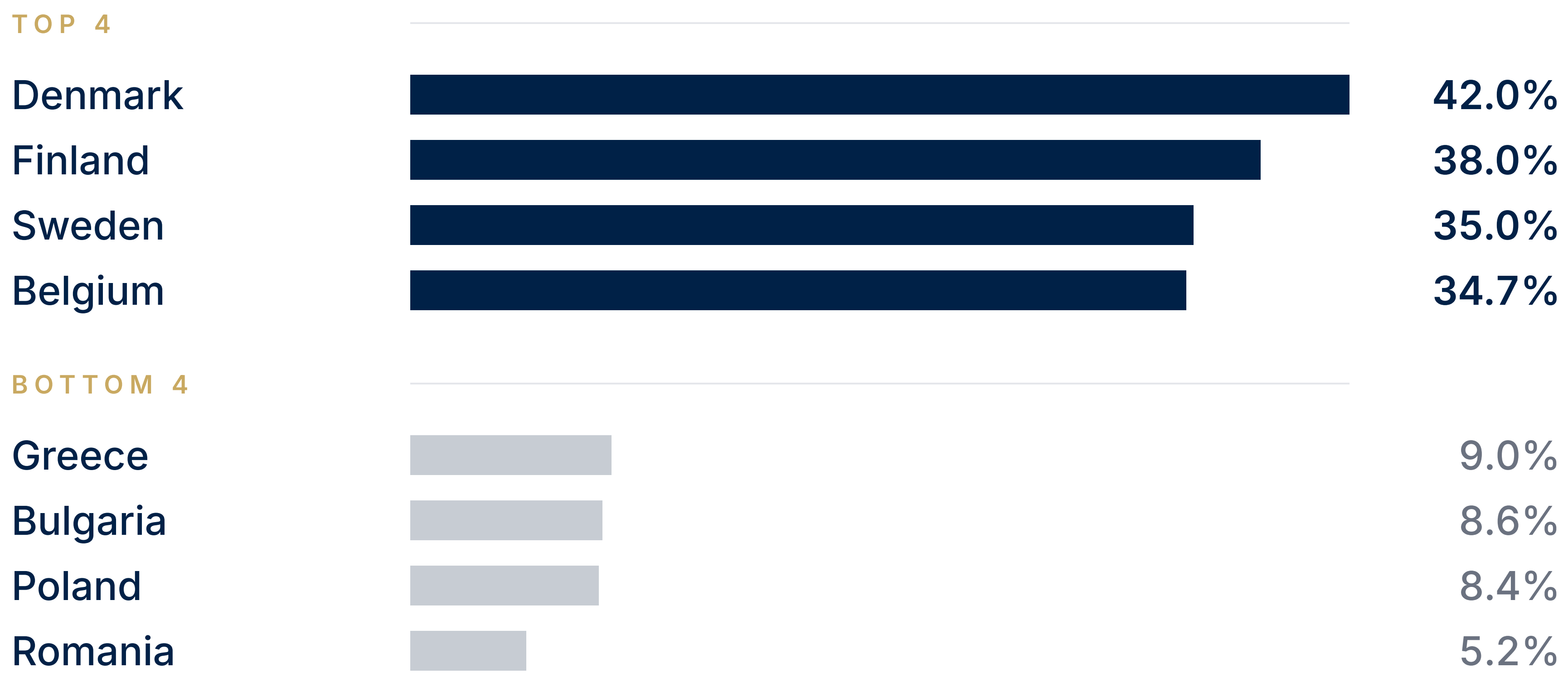
AI adoption tripled in four years — but the gap widened.



EU-27 mean rose from 7.7% (2021) to 20.7% (2025). Country range expanded from 22 pp to 37 pp over the same period.

Source: Eurostat (isoc_eb_ai).

Eight to one



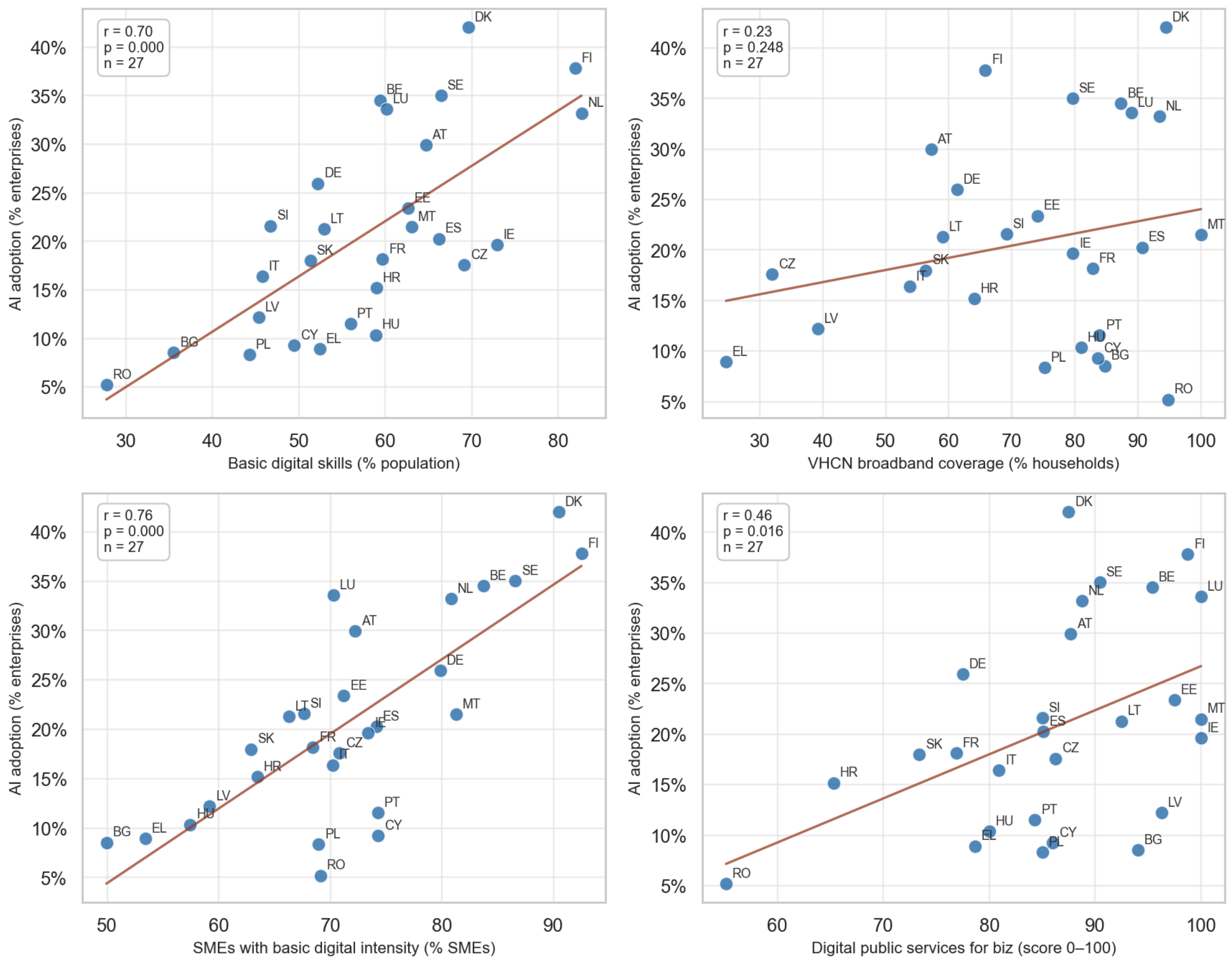
i The **Nordic–Benelux cluster** leads decisively. The bottom four are all from Eastern and Southern Europe.

ii The Denmark / Romania ratio is approximately **8×** — comparable to the per-capita income gap of the 1990s, before convergence.

iii In **relative** terms the gap has narrowed since 2021. In **absolute** terms it has widened by **15 percentage points**.

It's not the broadband — it's the firms and the people.

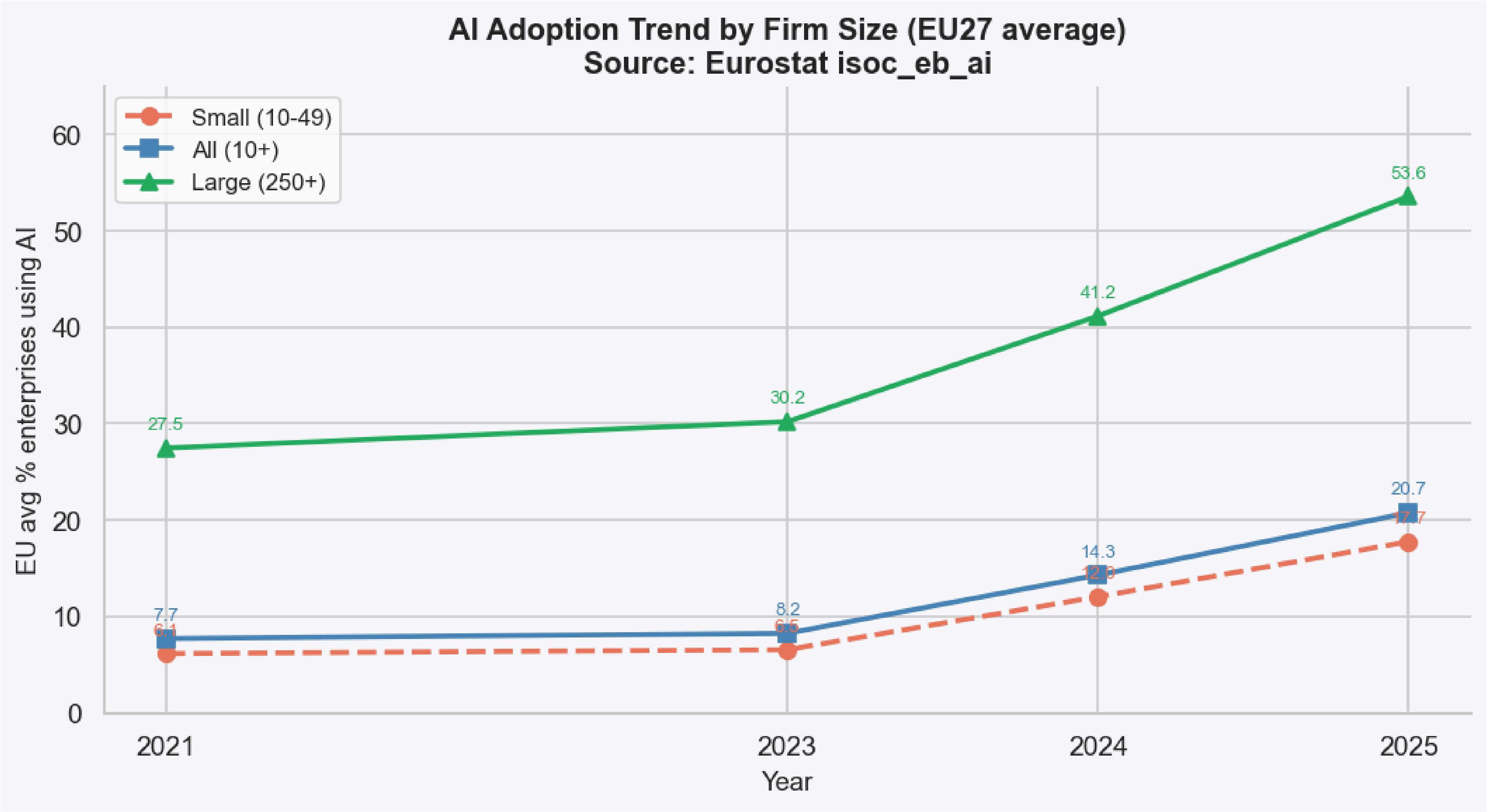
AI adoption vs Digital Decade readiness — EU-27, 2025



PREDICTOR	R	SIG.
SME digital intensity	0.76	***
Basic digital skills	0.70	***
Digital public services for business	0.46	*
VHCN broadband coverage	0.23	ns

Broadband infrastructure has saturated as a binding constraint. The bottleneck has moved upstream.

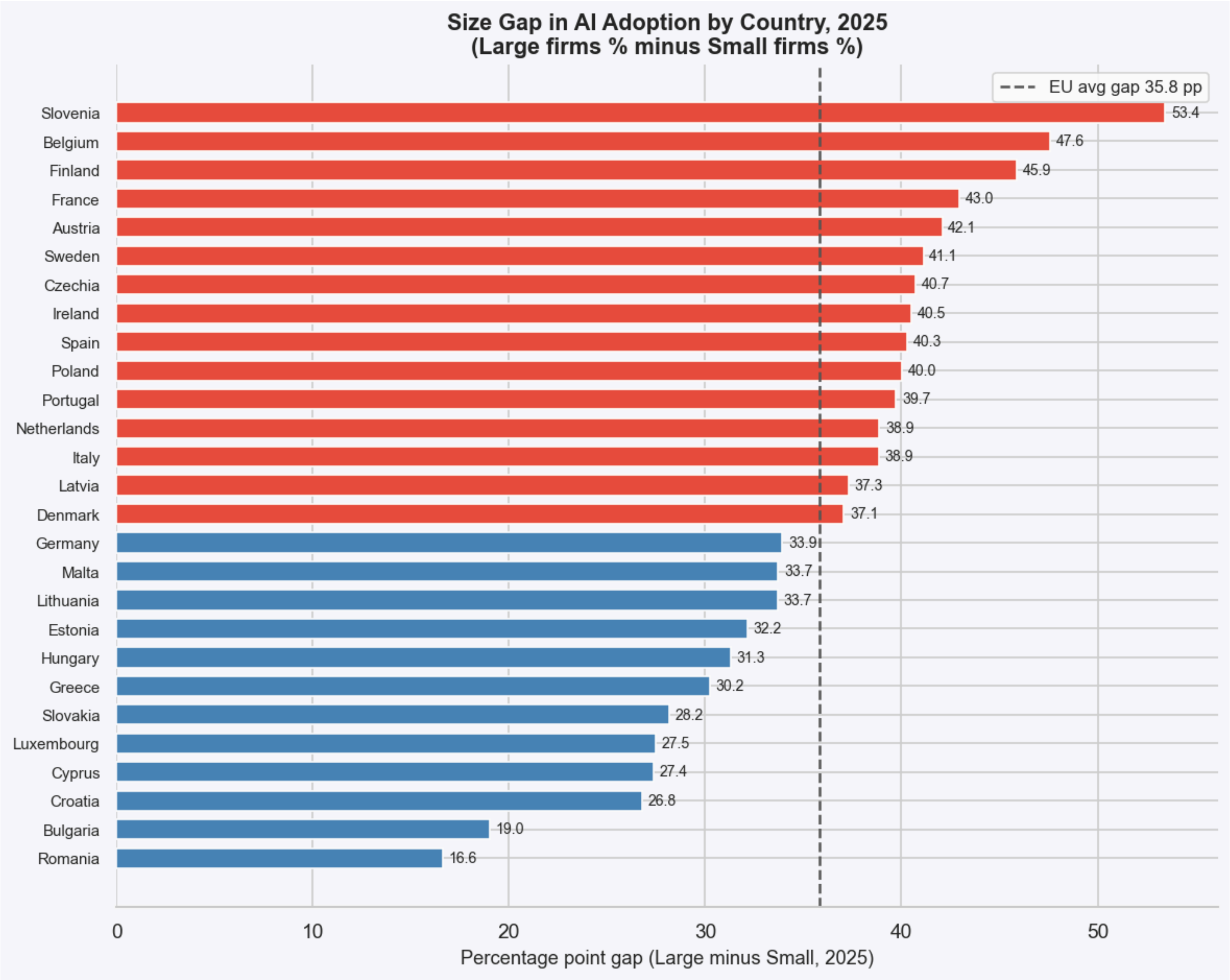
Large firms led the wave. Small firms followed two years later.



In 2025 large firms reach 54%, medium 30%, small 18%. The technology enters through large firms and diffuses downward.

Source: Eurostat (isoc_eb_ai).

Some economies have a small-firm problem, not a country problem.



PATTERN A

Dual economies

Slovenia · Belgium · France · Austria

Large firms have caught up. SMEs lag behind.

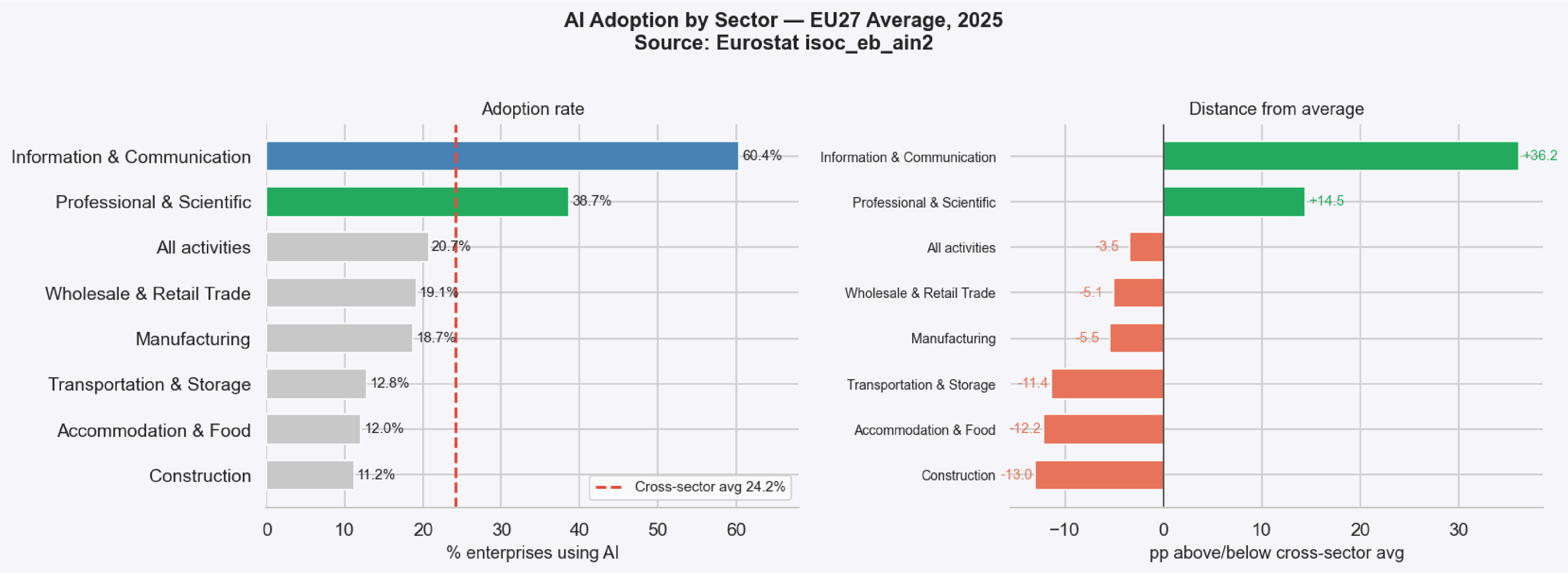
PATTERN B

Uniformly lagging

Romania · Bulgaria

Both segments below EU average. The size gap is small because the ceiling is low.

Three speeds: a digital frontier, a broad middle, a physical tail.



I.

Frontier >35%

ICT · Professional services

II.

Middle 15–25%

Manufacturing · Retail · Real estate

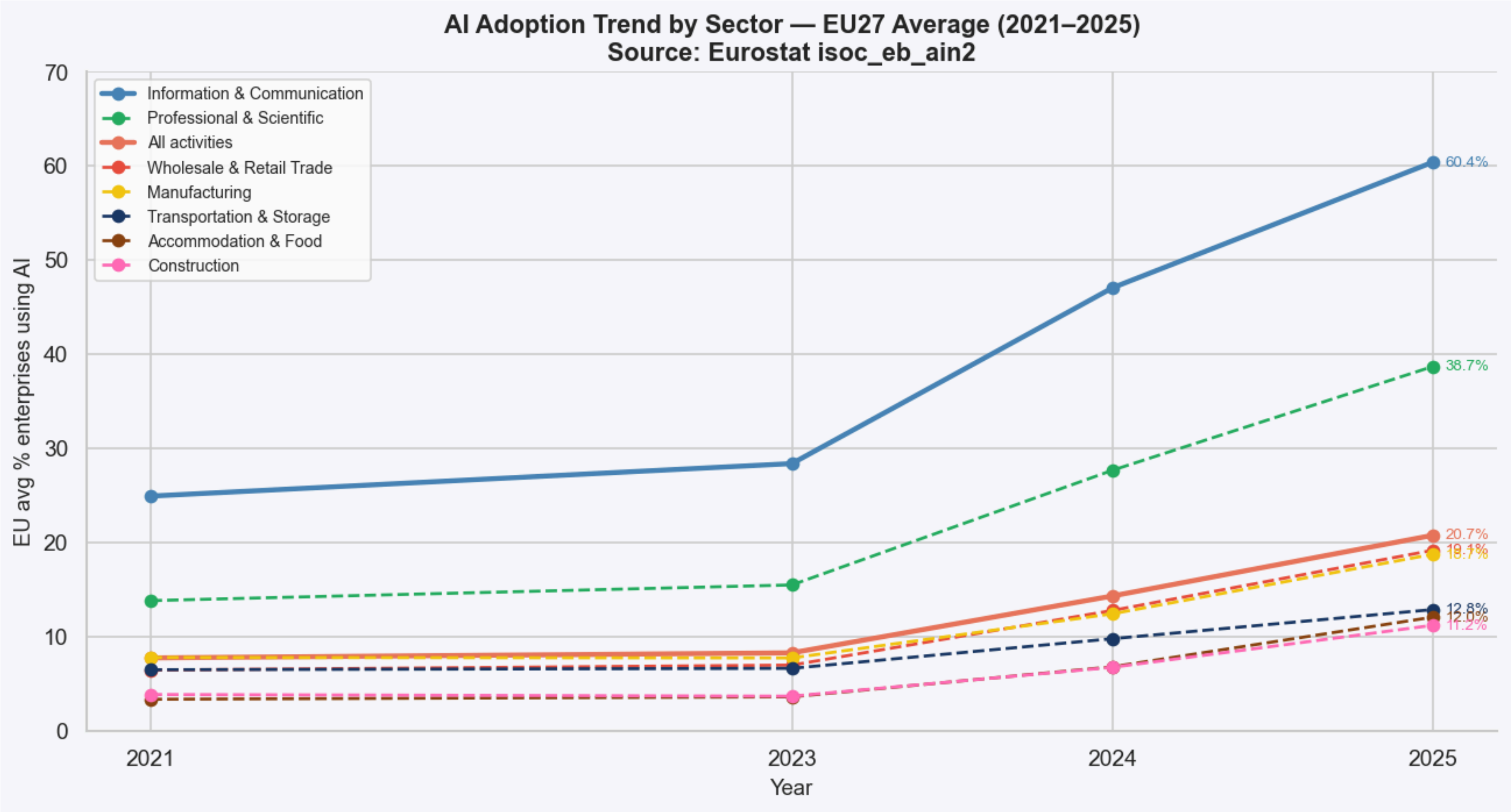
III.

Tail <15%

Transport · Hospitality · Construction

Source: Eurostat (isoc_eb_ain2). EU-27 average, 2025.

The leaders are pulling away — and the absolute gap widens every year.



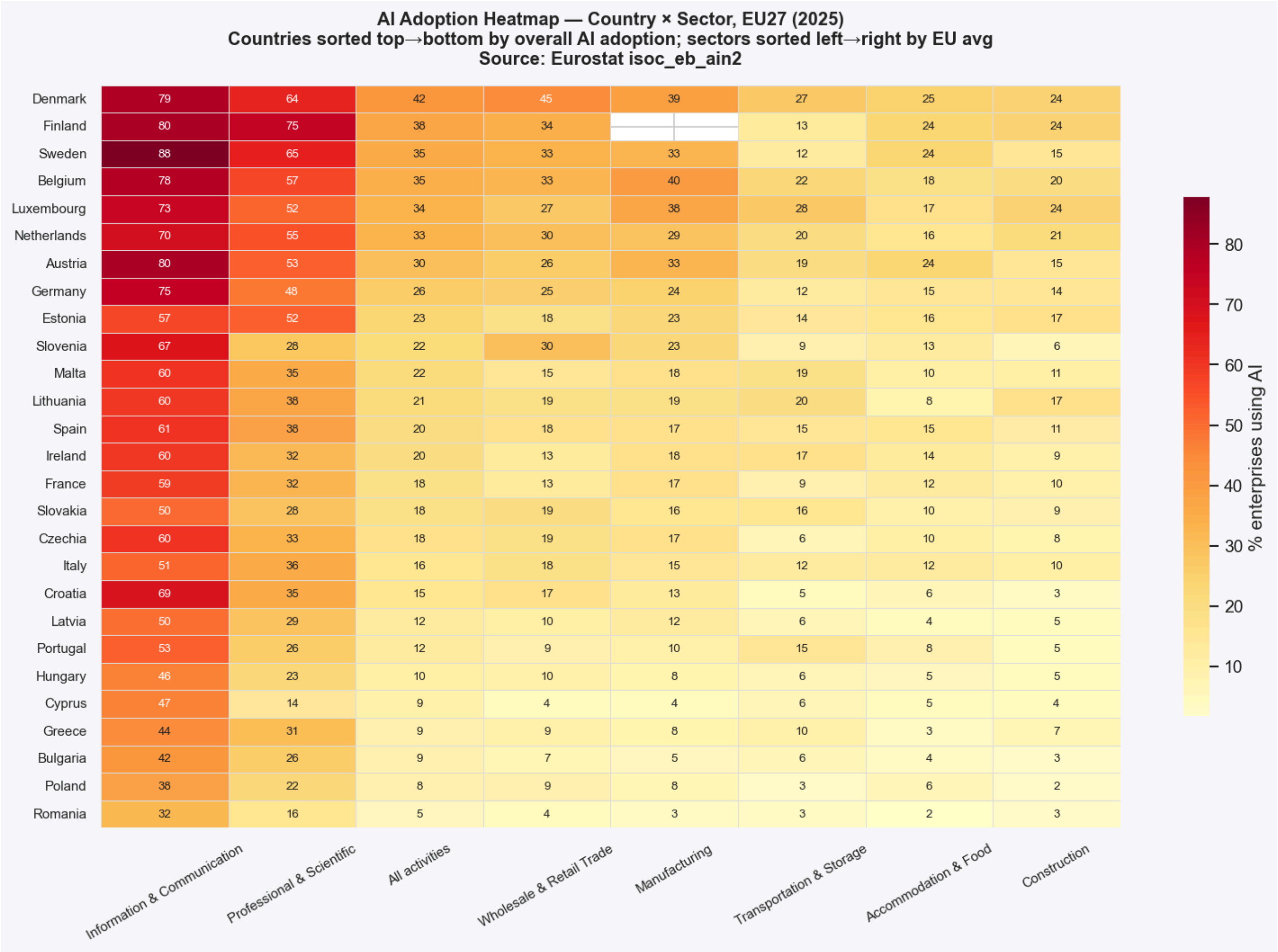
THE FRONTIER

ICT grew from 25% (2021) to **60%** (2025). **Professional services** tripled, from 14% to **39%**.

THE TAIL

Construction CAGR is **+31%**, faster than ICT's +25%. But starting from a near-zero base, the absolute gap keeps widening.

Country and sector both matter — and they combine in non-obvious ways.



A **Romanian ICT firm (32%)** *adopts AI less than a* **Belgian manufacturing firm (40%)**. *The leading sector of the laggard country is overtaken by the middle sector of the leader country.*

AI adoption is a multi-level phenomenon

LAYER I

Country

Digital maturity of firms (r = 0.76); skills of population (r = 0.70)

EVIDENCE

8× gap between Denmark and Romania. VHCN broadband does **not** predict adoption.

LAYER II

Firm size

Scale, in-house IT, capacity to absorb risk

EVIDENCE

Large firms **3×** more likely than small. Dual economies where SMEs lag despite strong large firms.

LAYER III

Sector

Task codifiability; complementary intangibles

EVIDENCE

ICT **60%** vs Construction **11%**. Absolute gap widens every year.

None of the three layers, taken alone, explains the variance.

Two objectives, two policies

PATH A · SPEED

Maximise EU-wide adoption

- Invest where returns are immediate
- Reinforce ICT and professional services
- Implicit strategy of the Nordics
- Probably hits the 75% target by 2030

PATH B · EQUITY

Reduce inequality across layers

- Target SME digital intensity in dual economies
- Workforce skills programmes for laggard countries
- Sector-specific support for the physical-economy tail
  IoT, computer vision, process digitalisation
- The harder, slower path

The Digital Decade target is silent on which of these two objectives the EU is pursuing.

What the data say

01 Infrastructure is no longer the binding constraint.

The bottleneck has moved from pipes to firms and to people.

02 The European AI divide is at least as much a small-firm gap as a country gap.

Dual economies are at least as numerous as uniformly lagging ones.

03 The sectoral divide is widening in absolute terms.

Higher CAGR for laggards does not translate into convergence when leaders started years ahead.

The right policy depends on the right level of disaggregation — country, size, and sector taken together.

Thank you

Questions?

DATA SOURCES

Eurostat — *isoc_eb_ai*, *isoc_eb_ain2*
European Commission — Digital Decade dashboard

SELECTED REFERENCES

Brynjolfsson, Rock & Syverson (2021) — *AEJ: Macro*
Calvino et al. (2018) — *OECD STI*
Acemoglu et al. (2022) — *JOLE*
Full bibliography in the written report.